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Parallel fibre receptive fields of Purkinje cells and interneurons are climbing fibre-specific

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Abstract

In cats decerebrated at the intercollicular level, the cutaneous parallel fibre receptive fields of Purkinje cells, molecular layer interneurons and Golgi cells in the cerebellar C3 zone were delineated by natural stimulation of the skin during extracellular unitary recordings. The locations of these receptive fields were compared with the climbing fibre receptive field of the local Purkinje cell and with the receptive fields of other neurons located along a beam of parallel fibres. The parallel fibre receptive fields of these neurons were highly specific to the local climbing fibre receptive field. In Purkinje cells, the parallel fibre receptive fields were located outside the climbing fibre receptive field of the same cell. In contrast, the parallel fibre receptive fields of interneurons were similar to the receptive field of the locally terminating climbing fibres. In both types of neurons, the parallel fibre receptive fields were small and had distinct borders. The location on the skin of the parallel fibre receptive fields differed conspicuously between neighbouring Purkinje cells and between neighbouring interneurons along a beam as well as between Purkinje cells and interneurons in the same electrode tracks. The remarkable specificity between the parallel fibre receptive fields in Purkinje cells and interneurons and the receptive field of the local climbing fibre is most easily explained by different forms of parallel fibre synaptic plasticity.

Introduction

The unusual arrangement of synapses on the Purkinje cell, which is contacted by several hundred thousand parallel fibres (Harvey & Napper, 1991) but only one climbing fibre (Ito, 1984), led early to the idea that the Purkinje cell is a memory device in which information is stored by climbing fibre-induced plastic changes of parallel fibre synapses. Such plasticity was later induced experimentally by conjunctive activation of parallel fibres and climbing fibres, which resulted in long-term depression (LTD) of parallel fibre synapses (Ito *et al*., 1982; Ekerot & Kano, 1985). However, the role of LTD and other possible forms of parallel fibre synaptic plasticity in selecting the input that drives the Purkinje cells is not known, nor is it known if parallel fibre synapses on interneurons in the cerebellar cortex are subject to experience-dependent modifications.

The C3 zone of the cerebellar cortex controls skilled movements via the corticospinal and rubrospinal tracts. In this zone, both climbing fibres and mossy fibres are activated from a limited number (30–40) of cutaneous receptive fields, which differ in their proximal extent and the location of their sensitivity centres (Ekerot *et al*., 1991; Garwicz *et al*., 1998). Climbing fibres with the same receptive fields on the skin terminate in narrow (50–150 μm) longitudinal microzones. Different microzones exert a functionally specific regulation of the red nucleus (Ekerot *et al*., 1995) and the motor cortex (Jörntell & Ekerot, 1999). Cutaneously activated mossy fibres tend to respect the climbing fibre-defined microzonal organization so that the population of mossy fibres within a microzone are usually maximally activated from a similar skin area as the local climbing fibre (Garwicz *et al*., 1998). Mossy fibres activate granule cells which transmit mossy fibre information along their T-shaped axons (ascending axon and parallel fibre) to Purkinje cells and inhibitory interneurons at distances up to 3.5 mm from the bifurcation site (Brand *et al*., 1976). Morphologically, virtually all parallel fibres passing through the dendritic tree of a Purkinje cell make synaptic contact with the cell (Harvey & Napper, 1991; Pitchitpornchai *et al*., 1994).

Because of the strict geometrical arrangement of the cerebellar cortex, neurons located at short distances along a beam of parallel fibres receive synapses from essentially the same population of parallel fibres. Hence, if all synapses have similar or random weights, the parallel fibre receptive fields of all neurons along a beam should be similar to the average receptive field of the mossy fibre population terminating beneath the parallel fibre beam. Accordingly, in single neurons, the parallel fibre receptive fields should not have any specific relationship to the local climbing fibre receptive field. The present paper aims to analyse whether the above organization does indeed apply or if the distribution of synaptic weights of parallel fibre synapses follows any other organization. The analysis was performed in the forelimb area of the cerebellar C3 zone, in which there is a fine topographical organization of mossy fibres and climbing fibres, which are strongly activated from small, specific parts of the skin (Ekerot *et al*., 1987; Garwicz *et al*., 1998). Because this system is now characterized at a relatively detailed level, we believe that it offers a reliable functional framework (see above) within which the principles of distribution of parallel fibre input to Purkinje cells and/or interneurons can be identified.

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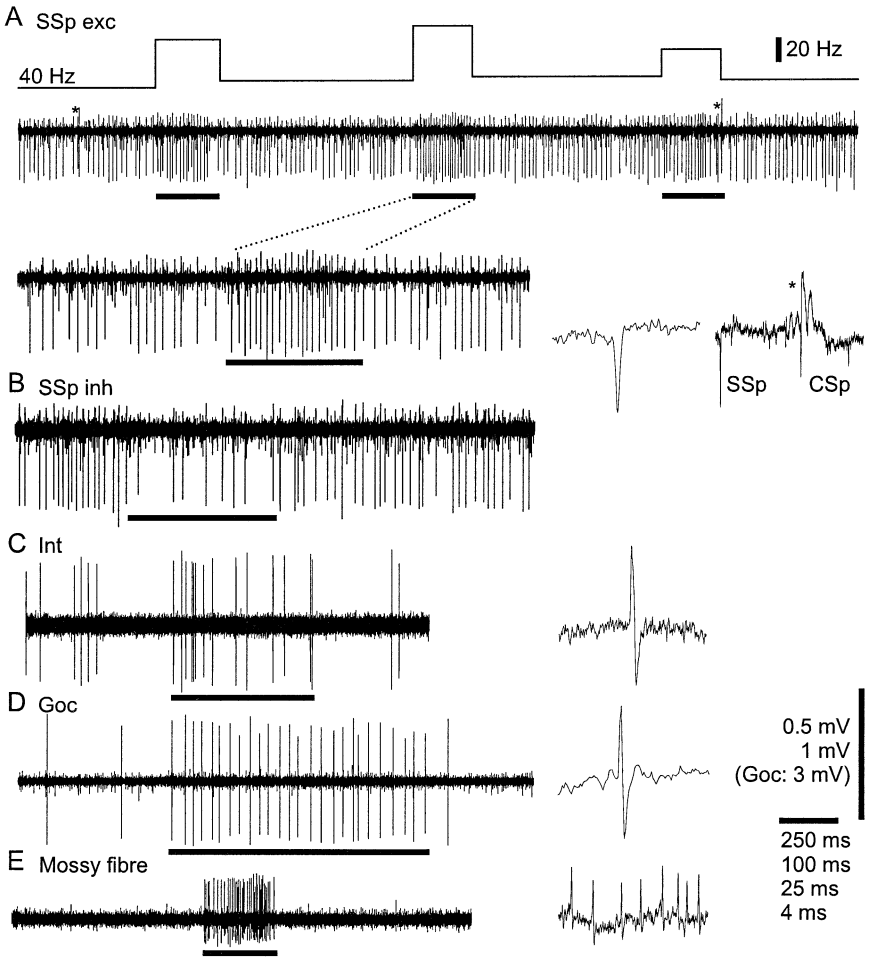

FIG. 1. Peripheral activation of cerebellar cortical neurons. (A) Simple spike (SSp) recording from a Purkinje cell during stroke across the excitatory cutaneous receptive field (approximate time of cutaneous contact indicated by horizontal line). Upper panels show the SSp response to three stimulations during a 3.6-s-long period both as raw data and as averaged firing frequencies for periods between and during stimulation. Lower panel shows the most vigorous response at higher sweep speed. Inset to the right: SSp and complex spike (CSp) of the cell is shown at a higher sweep speed to the right; the SSp is shown with a further expanded time-base to the left. (B) Same cell as in (A) but trace was recorded during stroke across inhibitory receptive field. CSp activation is indicated by asterisks in A and B. (C) Recording from putative interneuron (Int) located in the upper part of the molecular layer during stroke across the receptive field. Single spike shown to the right. (D) Recording from putative Golgi cell (Goc) during tonic light touch within the receptive field. Sweep to the right shows single spike. The spike could be recorded across some 0.1 mm of microelectrode travel. (E) Recording from mossy fibre terminal during rapid stroke across the receptive field. In A–E, time calibrations of 250 ms apply to the upper sweep, 100 ms apply to all other sweeps to the left, 25 ms apply to sweep displaying a SSp and a CSp and 4 ms apply to all other. 1 mV amplitude calibration applies to all sweeps except the uppermost and the recording from the Golgi cell. A rolling average spanning 2 ms (eight samples in the digitized data) has been subtracted from sweeps with long durations to reduce baseline shifts. This procedure did not affect the spike amplitudes.

Materials and methods

Nine adult cats were anaesthetized with propofol (Diprivan ® Zeneca Ltd, Macclesfield, Cheshire, UK) administered i.v., supplemented as necessary, and respirated with a mixture of N_2O (75–80%) and O_2 (20–25%). One animal was anaesthetized with thiopental (Pentothal ® Abbott Laboratories, North Chicago, Il 60064, USA) administered and supplemented as necessary i.v. The trachea was cannulated to allow artificial respiration. The femoral artery and vein were also cannulated. The end-expiratory CO_2, blood pressure and rectal temperature were maintained within physiological limits and a continuous infusion of glucose in Ringer–acetate was given. All wound areas were infiltrated with lidocaine (Xylocain ® Astra Södertälje, Sweden). After ≈ 1 h of maintained anaesthesia the left occipital lobe was resected and the animal was decerebrated; both hemispheres isolated with a cut at the level of the superior colliculus. The anaesthesia was then discontinued. For animals under propofol anaesthesia, the breathing gases was switched to 25% O_2 in air, a mixture used from the start for the animal under thiopental. The bony tentorium over the left cerebellar anterior lobe was removed. During anaesthesia and after decerebration, the animals had constricted pupils and stable mean blood pressure both at rest and during noxious stimulation. The EEG, measured after decerebration, from the intact (right) hemisphere, was characterized by periods of low-frequency, large-amplitude waves. To increase the mechanical stability of the brain, cerebrospinal fluid was drained through a hole in the dura made above the first vertebra. The lower cervical and upper lumbal spines were clamped, the animal was mounted in a stereotaxic frame with bars in the ears and mouth and kept paralysed with pancuronium (Pavulon ® Organon Teknika B.V., Boxtel, Holland). A pneumothor-

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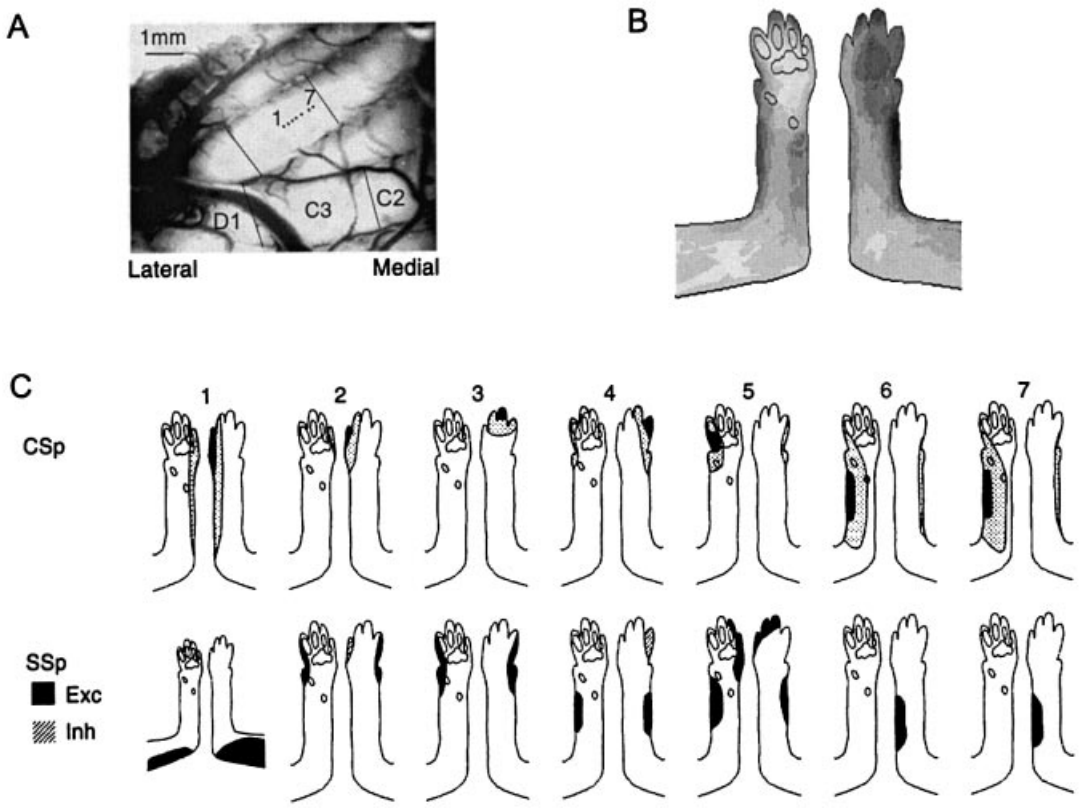

FIG. 2. Sample receptive fields of Purkinje cells along a beam of parallel fibres. (A) Dorsal view of left cerebellar anterior lobe (lobule V). The borders of the C3 zone (see Materials and methods) are indicated by lines. 1–7, location of electrode tracks for Purkinje cells shown in C. Rostral is upwards. (B) Superimposed receptive fields of 226 mossy fibre terminals in the forelimb area of the C3 zone (constructed from the data of Garwicz *et al*., 1998). The degree of shading corresponds to the degree of overlap relative to maximal overlap. Black indicates area of maximal overlap. (C) Peripheral receptive fields of complex spikes and simple spikes in Purkinje cells located along the same parallel fibre beam. CSp receptive fields are shown in the upper row and SSp receptive fields in the lower. Centre of maximal sensitivity for CSp receptive field is indicated in black and periphery in stippling. SSp excitatory and inhibitory receptive fields are black and hatched, respectively. In this and the following figures, pairs of limb outlines are used to indicate the location of receptive fields on the ventral and dorsal sides of the limb.

ax was also made to further increase mechanical stability. The anaesthesia was allowed to wear off for at least 6 h before microelectrode recording.

In the beginning of experiments, a freely movable surface electrode was used to record the cerebellar potentials evoked from various parts of the skin by brief electrical shocks through percutaneous needle electrodes. This allowed a fast delineation of the borders between the C3 zone and the neighbouring C2 and D1 zones (Fig. 2A; see Ekerot & Larson, 1979). This mapping also allowed a monitoring of the transmission in the very sensitive mossy fibre–granule cell–parallel fibre system (see Garwicz & Andersson, 1992, for extensive discussion about the sensitivity of this system), which are indicated by the amplitude of the N3 field potential (Eccles *et al*., 1967). Data collection did not start until this potential could be found at amplitudes exceeding 0.3 mV. The size of the N3 field potential was continuously monitored during microelectrode tracking, and data collection was discontinued as soon as there was obvious deterioration in this transmission.

A coiled silver wire was put on the surface of the C3 zone to stabilize the recording area. Tungsten-in-glass microelectrodes (exposed tip 3–20 μm) were inserted perpendicular to the cerebellar cortical surface and advanced in order to isolate a neuronal unit in the superficial cortical sheet. Tracks were separated by some 150 μm along the long axis of the folium. During tracking, the appearance of the evoked activity and the microdriver step counter was used to identify the layer recorded from. Once a unit was isolated, the entire body surface was investigated with manual stimulation in search of a cutaneous receptive field. The spike activity of the unit was continuously monitored through loudspeakers, by display on an oscilloscope and on a computer screen. In some cases, the data were digitzed at 40 kHz and stored on a computer disc by home-made software. During the mapping of receptive fields, we generally used light manual stimulation to get as specific stimulation as possible. To further avoid contaminating input from unstimulated parts of the skin or from joint movements caused by the stimulation, we always mapped receptive fields by supporting the forelimb in several different ways and by keeping it in a variety of positions. In each position, all parts of the forelimb skin were stimulated many times. From the skin area that was defined as belonging to the receptive field, distinct increases in discharge that were readily identifiable both audibly and visually were consistently evoked on manual stimulation. Such increases in discharge were not evoked from any other part of the body. When the receptive fields of simple spike (SSp) inhibition were investigated, carefully controlled stimulation was often required to avoid activation of the complex spike (CSp) and hence noninterneuronal inhibition of the SSp (Rawson & Tilokskulchai, 1981). In most cells, it was possible to selectively inhibit the SSp activity without activating the CSp by using a very light touch on the

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skin, at least in the majority of stimulations. In a few cells, the CSp was so sensitive to the peripheral stimulation that the presence of interneuronal SSp inhibition could not be determined. The receptive field mappings lasted for a minimum of 15–30 min for Purkinje cells and 5–15 min for interneurons, but in many cases we assured that the receptive fields stayed stable for about an hour or more. In different tracks, the order in which the local climbing fibre receptive field and the receptive fields of interneurons or SSp was mapped varied, so the relationship found between these types of receptive fields cannot be the result of climbing fibre-induced synaptic plasticity induced during the mapping.

The experimental procedures were approved in advance by the local Swedish Ethics Committee.

Results

Peripheral activation of cerebellar cortical neurons

We recorded from 127 Purkinje cells, 46 putative molecular layer interneurons and 15 putative Golgi cells located in the forelimb area of the C3 zone. The receptive fields of these cells were delineated by natural stimulation of the skin (touch, light stroke and light tapping) during extracellular unitary recordings. All Purkinje cells analysed discharged SSp, which reflect activation via parallel fibres, and CSp, which reflect activation via climbing fibres (Fig. 1A; see Ito, 1984). Both SSp and CSp occurred spontaneously at irregular rates, but stimulation of appropriate parts of the skin (i.e. the receptive field of the respective spike type) consistently gave rise to an increased discharge that was distinct also in single trials (Fig. 1A). Inhibition of SSp discharge (Fig. 1B) could also be evoked from specific parts of the skin. The properties of the CSp receptive fields were similar to those described for the climbing fibre input to the C3 zone in previous studies (Ekerot *et al*., 1987; Ekerot *et al*., 1991). The SSp receptive fields were generally similar to those of CSp in having small sizes and distinct borders. However, unlike CSp receptive fields, SSp receptive fields were unresponsive to noxious pinch, a property they share with the mossy fibres terminating in the same zone (Garwicz *et al*., 1998). Note that all receptive fields for excitation and inhibition of SSp discharge accounted for here refer to events during which no CSp occurred.

Putative interneurons in the molecular layer were characterized by the absence of spontaneous and driven climbing fibre responses (Fig. 1C). The Golgi cells recorded from had typical slow spikes and low maximal firing rates on peripheral activation (Fig. 1D), which clearly distinguished them from the other type of unit encountered in the granule cell layer, the mossy fibre terminal (Fig. 1E; see van Kan *et al*., 1993).

Parallel fibre activation of Purkinje cells with identified climbing fibre input

Figure 2A shows the location in the C3 zone of electrode tracks made to record from the seven Purkinje cells illustrated in Fig. 2C. The distances between the cells were ≈150 μm. The cutaneous input to the population of parallel fibres, that are responsible for all the cutaneously driven synaptic contacts that are made on these cells, should largely correspond to that shown in Fig. 2B. This figure represents the average receptive field of the mossy fibres terminating in the C3 zone (Garwicz *et al*., 1998). (Note, though, that the contribution from the mossy fibre input to the adjacent C2 and D1 zones is not included).

Conspicuously, the excitatory SSp receptive fields were small, distinct and differed dramatically between the Purkinje cells (Fig. 2C). The SSp receptive field never overlapped or resembled the CSp receptive field, with the single exception of a partial overlap (see Fig. 3C). Moreover, the SSp receptive fields remained stable also throughout long recording times (up to ≈2 h) despite frequently occurring excitability changes of the Purkinje cell as monitored by the level of spontaneous and evoked SSp firing. Interestingly, the SSp receptive fields of individual Purkinje cells seemed to correspond to one or sometimes two (see cell 5 in Fig. 2C) of the more than 30 types of cutaneous forelimb receptive fields found for mossy fibres in the C3 zone (Garwicz *et al*., 1998).

SSp receptive fields were similar in Purkinje cells with identical or nearly identical CSp receptive fields (Figs 2 and 3). In the total material, most Purkinje cells (83%) were associated with a single distinct SSp receptive field. In 17% of the Purkinje cells, the SSp receptive field included two distinct components to create a double receptive field (hence the higher number of SSp than CSp receptive fields in Fig. 3B–D). The lower row in Fig. 3 shows the receptive fields for mossy fibres (data from Garwicz *et al*., 1998) found beneath the same types of CSp receptive fields as in the top row. Note the virtually complete absence of overlap between SSp and mossy fibre receptive fields for corresponding CSp receptive fields in Fig. 3A–C.

Inhibitory receptive fields of Purkinje cells and receptive fields of interneurons

Inhibitory receptive fields were found in 36% of the Purkinje cells. In 93% of these cells the spontaneous SSp activity was exclusively inhibited from the CSp receptive field, in many cases only from its central part (see Figs 2 and 4). The remaining 7% had double inhibitory receptive fields, one located in the CSp receptive field and one elsewhere on the forelimb. The spatial characteristics of the inhibitory receptive fields suggested that stellate and basket cells have peripheral receptive fields similar to those of locally terminating climbing fibres. This notion was supported by recordings from putative interneurons in the molecular layer. The receptive fields of these interneurons were nearly identical with the CSp and inhibitory receptive fields of the local Purkinje cell (Fig. 4A and B). Furthermore, Golgi cells in the same electrode tracks also had similar receptive fields (Fig. 4B). In general, the receptive fields of interneurons and SSp inhibition were located within the borders of the CSp receptive field of the track, and they were often confined to or focused on the receptive field centre of the CSp. As a result, when the receptive fields of SSp inhibition and interneurons found in tracks with similar CSp receptive fields were pooled, they were found to be sharply delimited (Fig. 4C and D). Note also that the pooled receptive field of the mossy fibres recorded in the same microzones in a previous study (Garwicz *et al*., 1998) were centred around the receptive fields of the CSp, SSp inhibition and molecular layer interneurons, but had a wider distribution on the skin (Fig. 4C and D).

In all interneurons encountered, an excitatory receptive field that provided strong input was easily identified. The failure to identify inhibitory receptive fields in about two thirds of the Purkinje cells (cf. above) could only to some extent be explained by factors such as a low spontaneous SSp activity, making it difficult to discern inhibitory effects, or difficulties in separating interneuronal inhibition from CSp-induced depression of SSp activity because of the CSp being very sensitive to peripheral activation.

Discussion

Here we have shown that both Purkinje cells and putative inhibitory interneurons in the C3 zone have small, well defined parallel fibre

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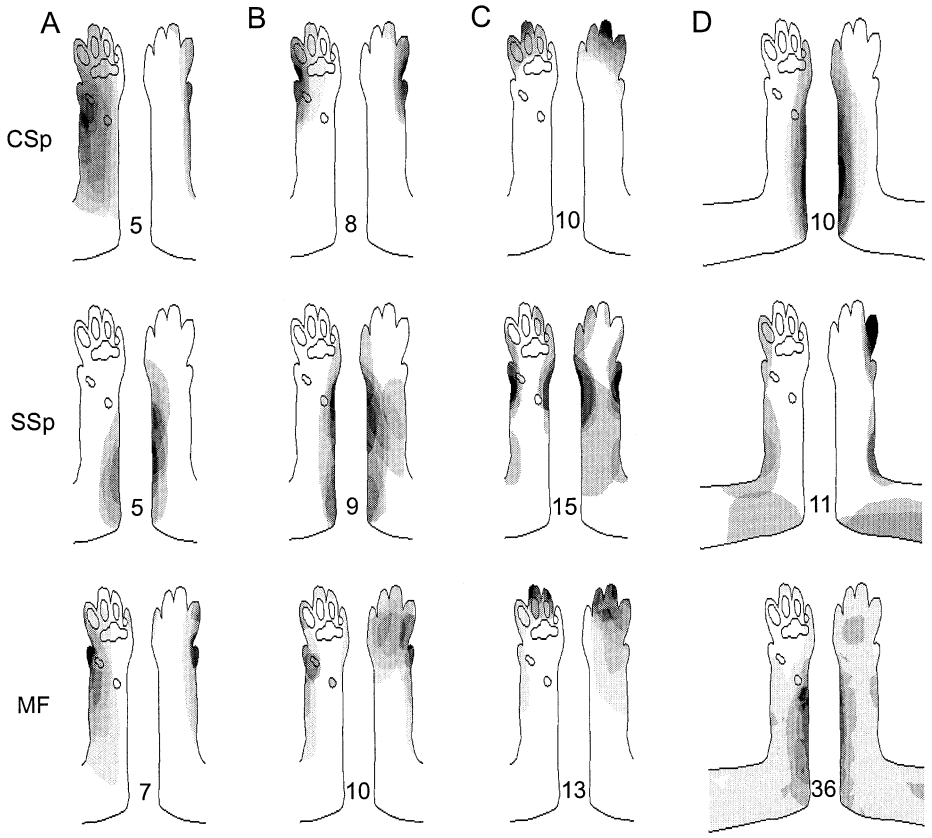

FIG. 3. Correlation between CSp and SSp receptive fields. (A) Purkinje cells with CSp receptive field centre at the ventral base of the first digit and with CSp receptive field periphery extending up on the ventral side of the forearm were pooled (receptive field corresponding to class 5b, cf. Garwicz *et al*., 1998). Their CSp (top) and SSp (middle) receptive fields were superimposed. The lower panel is based on previous data (Garwicz *et al*., 1998) and indicates the mossy fibre (MF) receptive fields found in tracks with the same climbing fibre receptive fields as in the top panel. (B) Display as in (A) but for Purkinje cells with CSp receptive field centres located along the first digit and the radial part of the second digit (class 2a). (C) Display as in A but CSp receptive field centres were located on the third digit (class 1c). Note that the SSp receptive field on the third digit was part of a double receptive field. (D) Display as in A but CSp receptive field centres were located along the ulnar part of the forearm (class 6c). In A–D, the level of shading indicates the degree of overlap, with black representing maximal overlap. For CSp receptive fields, sensitivity centres have double weight relative to the peripheral parts of the receptive fields (see Garwicz *et al*., 1998 for extensive description of this superimposing method). Numbers indicate the number of superimposed receptive fields.

receptive fields that are specific for the local climbing fibre input. In Purkinje cells, the cutaneous parallel fibre input mainly represents receptive fields found only for mossy fibres terminating outside their microzone whereas the parallel fibre receptive fields of inhibitory interneurons generally was of a type which most often is found in mossy fibres terminating in the same microzone.

Excitatory cutaneous parallel fibre receptive fields in Purkinje cells

The relationship between SSp and CSp receptive fields found in the present study seems to be at odds with the conclusions drawn in two earlier studies of the afferent input to Purkinje cells (Eccles *et al*., 1972; Bower & Woolston, 1983). Eccles *et al*. (1972), in an investigation in the cat in which the C3 zone in the anterior lobe seems to have been partly covered, found that tapping to one part of a limb often evoked a combined SSp and CSp response in single Purkinje cells (referred to as MF- and CF-evoked responses in that paper). However, the taps were given to paws, which were fixed in moulds, an approach which in our experience can be misleading. For example, we regularly found that tapping the dorsal side of a limb that rests against the ground with its ventral side is very likely to falsely give the impression that the unit recorded from has a dorsal receptive field when its receptive field actually is located fully or partly on the ventral side. Bower & Woolston (1983) studied the receptive fields of single Purkinje cells and compared them with the receptive fields of the evoked activity in the underlying granule cell layer in Crus I and II of the posterior cerebellar lobe in the rat. In contrast to our results, they found that the strongest excitatory parallel fibre input to Purkinje cells was from the cutaneous area, which most strongly excited the underlying granule layer. Possible explanations for the differences are true differences between different functional areas of the cerebellar cortex or differences in experimental conditions. For example, it is remarkable that Bower and Woolston found the same results whether the preparation was mainly, partly or not at all under the influence of sodium pentobarbital. Barbiturates are well known to depress the mossy fibre–Purkinje cell transmission (Körlin & Larson, 1970; Latham & Paul, 1971; Gordon *et al*., 1973). In one experiment in the present study, we noted a total or near total absence of peripheral activation of SSp activity and of N3 field potentials evoked by electrical stimulation of the skin up to 15 h after discontinuation of a one-and-a-half hour long anaesthesia under the short-acting barbiturate thiopenthal. With propofol as the anaesthetic agent before decerebration, SSp activity was usually easily driven by peripheral stimulation and the evoked N3 field potentials had maximal amplitudes in the range of 0.5–1 mV.

Because the excitatory SSp receptive fields were so specific to the climbing fibre receptive field recorded in the same cell, one of the implications of the present findings is that SSp receptive fields also have a microzonal organization. This microzonal organization would follow the same boundaries as the climbing fibre receptive fields (Ekerot *et al*., 1991), but have a distinctly different somatotopy.

Receptive fields of cerebellar interneurons

To our knowledge the present paper gives the first systematic description of cutaneous receptive fields of stellate and basket cells. These inhibitory interneurons were identified by their location in the molecular layer and absence of evoked and spontaneous climbing fibre responses. Their excitatory parallel fibre receptive fields differed from those of Purkinje cells in that they were, with few exceptions, confined to the receptive field of locally terminating climbing fibre. Hence, the receptive fields of interneurons were also nearly identical to the inhibitory receptive fields of Purkinje cells in the same microzone, a finding that gives additional support for their identifi-

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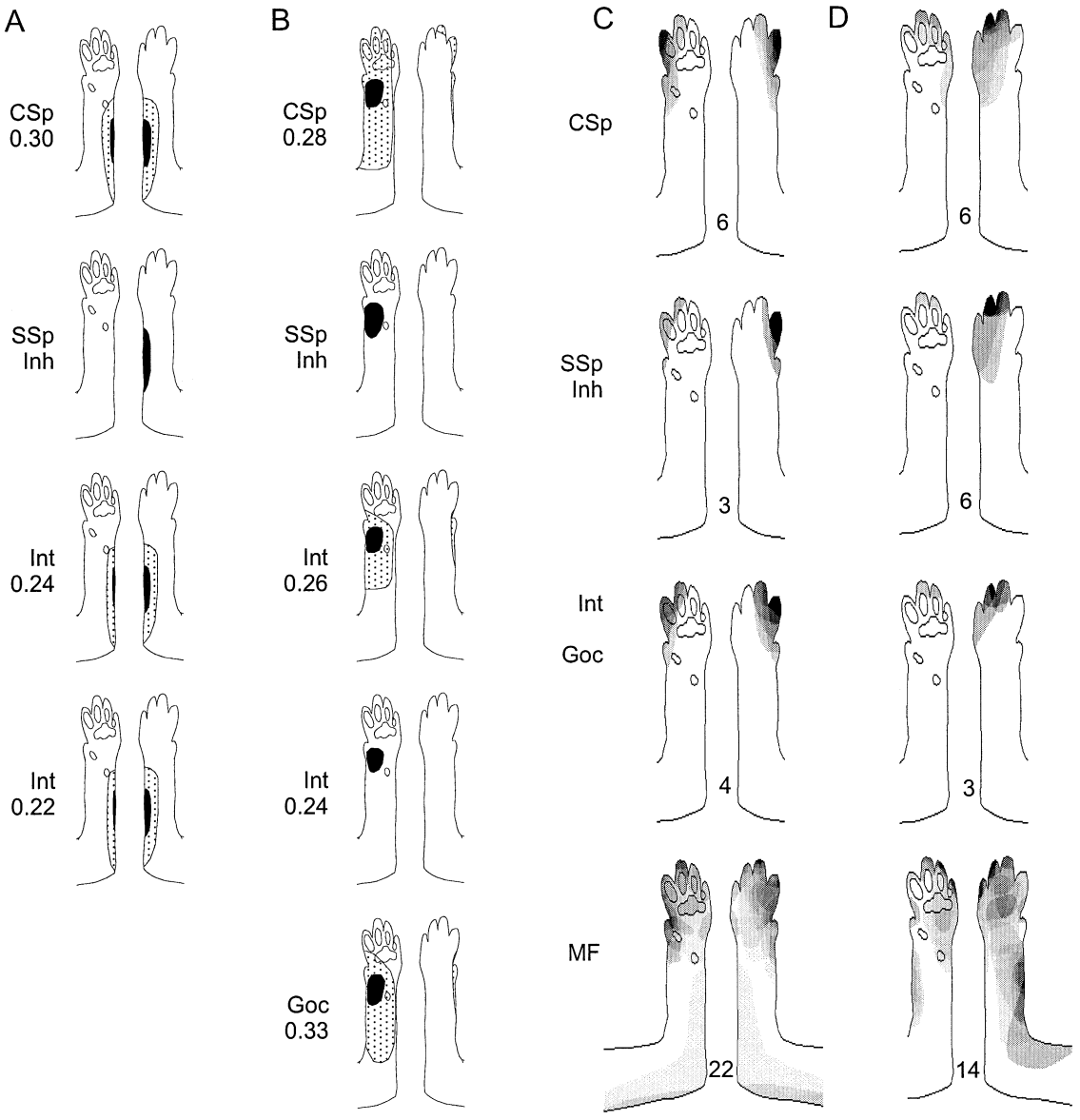

FIG. 4. Correlation between receptive fields of CSp, SSp inhibition and putative inhibitory interneurons. (A and B) The receptive fields of putative inhibitory interneurons and Purkinje cells in two sample tracks. Centres and peripheries of receptive fields are indicated in black and stippling, respectively. Numbers indicate recording depths in mm. From above: Purkinje cell CSp receptive field, Purkinje cell inhibitory receptive field for SSp, receptive fields of molecular layer interneurons and Golgi cells. (C and D) Purkinje cells with similar CSp receptive fields were pooled; their inhibitory SSp receptive fields and the receptive fields of interneurons encountered in the same track were superimposed separately (cf. Fig. 3). Also shown are the mossy fibre receptive fields found in tracks with corresponding CSp receptive fields in a previous study (constructed from the data of Garwicz *et al*., 1998). (C) Inhibitory and interneuronal receptive fields for Purkinje cells with CSp receptive field centres located on the tip of the second digit (corresponding to receptive field classes 1b and 2b, cf. Garwicz *et al*., 1998). Two out of four interneurons were putative Golgi cells. (D) Display as in C but for Purkinje cells with CSp receptive field centres located on the tip of the fourth digit (classes 1d and 2d). One out of three interneurons was a putative Golgi cell.

cation as inhibitory interneurons. These findings further suggest that the interneuronal inhibition of Purkinje cells is mainly restricted to neurons located within the same microzone.

Indirect evidence for climbing fibre-specific learning of the parallel fibre receptive fields in Purkinje cells and interneurons

There are four findings, each of which indicates that the excitatory parallel fibre receptive fields of Purkinje cells and interneurons are the result of a highly specific selection out of the vast pool of parallel fibre synapses that are available for each neuron.

(i) There were large differences in parallel fibre input between cells located at short distances (usually 150 μm) along a beam of parallel fibres, despite the fact that the change in total parallel fibre content along such a distance should only be ≈2.5% (assuming an average parallel fibre length of 6 mm; cf. Brand *et al*., 1976).

(ii) The parallel fibre receptive fields were different from the average receptive field of mossy fibres terminating in the C3 zone (Fig. 2B), which to a large part should correspond to the receptive field of the population of parallel fibres passing through the dendritic trees of single cells. In fact, the receptive fields of the parallel fibre input were small and resembled the receptive fields found for single mossy fibres (Garwicz *et al*., 1998). This indicates that the parallel fibre receptive fields described here represent specifically selected groups of parallel fibres with similar receptive fields.

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(iii) Purkinje cells and putative interneurons located in the same electrode track in the same cortical layer had completely different parallel fibre receptive fields despite the fact that the two types of neurons get their input from the same group of parallel fibres.

(iv) Purkinje cells with the same type of climbing fibre receptive fields were associated with one or a few parallel fibre receptive fields. Furthermore, the inhibitory receptive fields of Purkinje cells and excitatory receptive fields of putative interneurons in a microzone were usually confined to the receptive fields of locally terminating climbing fibres.

One can imagine two possible ways in which this climbing fibre-specific distribution of parallel fibre synaptic weights have been accomplished. One is that the synaptic weights of the different parallel fibre synapses are already specified when these synapses are created in early development and the other is that parallel fibre synaptic plasticity is entirely or partly responsible. In both cases the mechanisms behind the highly specific relationship between climbing fibre receptive fields and parallel fibre receptive fields in Purkinje cells and interneurons has to be explained. While we cannot of course in the present study totally dismiss the first possibility, we believe that the degree of specificity makes it highly likely that parallel fibre synaptic plasticity has had a major role in the establishment of these receptive fields. One cerebellar learning process, the previously described climbing fibre-induced LTD of parallel fibre synapses on Purkinje cells, is considered to have a role in the adaption of the vestibulo-ocular reflex in adult animals (Ito, 1984). An important question is then whether LTD can be responsible for the receptive fields of Purkinje cells in the present study. The absence of parallel fibre input from the climbing fibre receptive fields is a finding which is in line with the known properties of LTD. However, it is difficult to explain the generation of the parallel fibre receptive fields described here solely by a process leading to reduction of synaptic strength. First, the sharp borders and the stability of the receptive fields during spontaneous variations in excitability indicate an absence of the 'iceberg' effect that would be expected if the driving synapses were only selected by LTD (LTD would be expected to have a gradually weaker effect on parallel fibre synapses the further away their receptive fields are from the climbing fibre receptive field of the cell). Secondly, in the pooled data, a group of Purkinje cells with similar CSp receptive fields were often associated with two or three types of SSp receptive fields that were located on distinctly different parts of the limb. Multiple climbing fibre-specific SSp receptive fields seem impossible to reconcile with a situation in which LTD is the only mechanism for synaptic modification. Rather, it seems that the above observations can only be explained by assuming a novel learning mechanism, in which a small subset of the parallel fibre synapses on the Purkinje cell is selected to be potentiated.

For the putative inhibitory interneurons, the possibility that the parallel fibre synaptic weights are established during the creation of these synapses should be considered more carefully because the parallel fibre receptive fields of these neurons resemble those of the mossy fibre terminals in the underlying granule layer. Hence, an inherent radial organization of the granule cell–interneuron innervation, in which interneurons early in development are predetermined to be activated only by the ascending axons of granule cells, might appear feasible. However, because the receptive fields of these interneurons correspond only to the receptive fields of some of the locally terminating mossy fibres (Fig. 4C and D), this explanatory model does not suffice. Most probably, there must also be a learning mechanism. A surprising finding was that the parallel fibre receptive fields of interneurons typically were similar to that of the climbing fibre of a nearby Purkinje cell in the same track. Hence, the selection of driving parallel fibre synapses in interneurons may be specific to the local climbing fibre. If this is the case, it may imply a crucial role for the recently demonstrated collateral innervation of molecular layer interneurons and Golgi cells by climbing fibres, which innervate Purkinje cells located within the same parasagittal plane as the interneurons (Sugihara *et al*., 1999).

The majority of parallel fibre synapses are electrically silent

Morphological investigations have shown that virtually all parallel fibres that pass the dendritic tree of a Purkinje cell make synaptic contact with the cell (Harvey & Napper, 1991; Pitchitpornchai *et al*., 1994). The parallel fibre receptive fields of most Purkinje cells in the present investigation corresponded to one of the 30–40 types of cutaneous receptive fields found in mossy fibres. Hence, the effective parallel fibre input to a Purkinje cell can only correspond to a few per cent of the total number of cutaneously activated parallel fibres making synaptic contact with the cell. The spontaneous activity of the Purkinje cells and interneurons in our preparation (see Fig. 1) indicate that the membrane potential of these cells are normally around firing threshold and hence very little excitatory drive is required to produce a detectable difference in the firing rate. This suggests that the majority of cutaneously activated parallel fibre synapses on Purkinje cells and interneurons, i.e. those that do not drive the cell, not only are subjected to a down-regulation but are electrically silent. A similar estimation of the number of electrically effective synapses in Purkinje cells was recently made in a study using electrical stimulation of parallel fibres and photo-release of caged glutamate (Wang *et al*., 2000). Similarly, in hippocampus, a large proportion of excitatory synapses are normally electrically silent (Liao *et al*., 1995). Of course, it is uncertain whether all of the silent parallel fibre synapses have been subjected to LTD or if some are electrically silent from birth, as is the case in hippocampus (Durand *et al*., 1996), and have persisted in that state.

Concluding remarks

It may seem surprising that the parallel fibre receptive fields were so similar between animals and that most parallel fibre synapses do not seem to be used in the normal information processing. It may be speculated that the similarities in receptive fields reflect similarities in the movement capabilities of the different animals and that a vast pool of recruitable parallel fibre synapses are needed during the learning of new movement patterns. That the silent parallel fibre synapses are recruitable will be shown in a paper currently in preparation, in which electrical cutaneous stimulation is used to induce new small and specific receptive fields in Purkinje cells, without any obvious changes in the receptive fields of mossy fibres and interneurons.

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Abbreviations

CSp, complex spike; LTD, long-term depression; SSp, simple spike.

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